

## JOINS AND SET OPERATIONS

```
[amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 9.7.1 Homebrew

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

[mysql> USE join_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> CREATE TABLE class (
  ->     id INT,
  ->     name VARCHAR(30)
[  -> );
Query OK, 0 rows affected (0.009 sec)

mysql> CREATE TABLE class_info (
  ->     id INT,
  ->     address VARCHAR(30)
[  -> );
Query OK, 0 rows affected (0.008 sec)

mysql> INSERT INTO class VALUES
  -> (1,'abhi'),
  -> (2,'adam'),
[  -> (4,'alex');
Query OK, 3 rows affected (0.003 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO class_info VALUES
  -> (1,'DELHI'),
  -> (2,'MUMBAI'),
[  -> (3,'CHENNAI');
Query OK, 3 rows affected (0.003 sec)
Records: 3  Duplicates: 0  Warnings: 0

[mysql> SELECT * FROM class;
+-----+-----+
| id    | name |
+-----+-----+
| 1     | abhi |
| 2     | adam |
| 4     | alex |
+-----+-----+
3 rows in set (0.000 sec)
```

```

mysql> SELECT * FROM class_info;
+-----+-----+
| id | address |
+-----+-----+
| 1 | DELHI |
| 2 | MUMBAI |
| 3 | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> SELECT class.name, class_info.address
-> FROM class
[ -> CROSS JOIN class_info;
+-----+-----+
| name | address |
+-----+-----+
| alex | DELHI |
| adam | DELHI |
| abhi | DELHI |
| alex | MUMBAI |
| adam | MUMBAI |
| abhi | MUMBAI |
| alex | CHENNAI |
| adam | CHENNAI |
| abhi | CHENNAI |
+-----+-----+
9 rows in set (0.000 sec)

mysql> SELECT * FROM class;
+-----+-----+
| id | name |
+-----+-----+
| 1 | abhi |
| 2 | adam |
| 4 | alex |
+-----+-----+
3 rows in set (0.000 sec)

mysql> TRUNCATE TABLE class;
Query OK, 0 rows affected (0.010 sec)

mysql> INSERT INTO class VALUES
-> (1,'abhi'),
-> (2,'adam'),
-> (3,'alex'),
[ -> (4,'anu');
Query OK, 4 rows affected (0.002 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM class;
+-----+-----+
| id | name |
+-----+-----+
| 1 | abhi |
| 2 | adam |
| 3 | alex |
| 4 | anu |
+-----+-----+
4 rows in set (0.000 sec)

```

```

mysql> SELECT * FROM class_info;
+-----+-----+
| id | address |
+-----+-----+
| 1 | DELHI |
| 2 | MUMBAI |
| 3 | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> SELECT class.name, class_info.address
-> FROM class
-> INNER JOIN class_info
[ -> ON class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI |
| adam | MUMBAI |
| alex | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> SELECT class.name,
-> class_info.address
-> FROM class
-> INNER JOIN class_info
[ -> ON class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI |
| adam | MUMBAI |
| alex | CHENNAI |
+-----+-----+
3 rows in set (0.001 sec)

mysql> SELECT *
-> FROM class
[ -> NATURAL JOIN class_info;
+-----+-----+
| id | name | address |
+-----+-----+
| 1 | abhi | DELHI |
| 2 | adam | MUMBAI |
| 3 | alex | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> INSERT INTO class VALUES
[ -> (5, 'ashish');
Query OK, 1 row affected (0.002 sec)

mysql> INSERT INTO class_info VALUES
-> (7, 'NOIDA'),
[ -> (8, 'PANIPAT');
Query OK, 2 rows affected (0.001 sec)
Records: 2 Duplicates: 0 Warnings: 0

```

```
[mysql> SELECT * FROM class;
```

| id | name   |
|----|--------|
| 1  | abhi   |
| 2  | adam   |
| 3  | alex   |
| 4  | anu    |
| 5  | ashish |

```
5 rows in set (0.000 sec)
```

```
[mysql> SELECT * FROM class_info;
```

| id | address |
|----|---------|
| 1  | DELHI   |
| 2  | MUMBAI  |
| 3  | CHENNAI |
| 7  | NOIDA   |
| 8  | PANIPAT |

```
5 rows in set (0.000 sec)
```

```
mysql> SELECT class.name,  
->      class_info.address  
-> FROM class  
-> LEFT OUTER JOIN class_info  
-> ON class.id = class_info.id;
```

| name   | address |
|--------|---------|
| abhi   | DELHI   |
| adam   | MUMBAI  |
| alex   | CHENNAI |
| anu    | NULL    |
| ashish | NULL    |

```
5 rows in set (0.001 sec)
```

```
mysql> SELECT *  
-> FROM class  
-> LEFT OUTER JOIN class_info  
-> ON class.id = class_info.id;
```

| id | name   | id   | address |
|----|--------|------|---------|
| 1  | abhi   | 1    | DELHI   |
| 2  | adam   | 2    | MUMBAI  |
| 3  | alex   | 3    | CHENNAI |
| 4  | anu    | NULL | NULL    |
| 5  | ashish | NULL | NULL    |

```
5 rows in set (0.001 sec)
```



```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
[ -> WHERE class_info.id IS NULL;
```

| id | name   | id   | address |
|----|--------|------|---------|
| 4  | anu    | NULL | NULL    |
| 5  | ashish | NULL | NULL    |

2 rows in set (0.001 sec)

```
mysql> SELECT class.id, class.name
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
[ -> WHERE class_info.id IS NULL;
```

| id | name   |
|----|--------|
| 4  | anu    |
| 5  | ashish |

2 rows in set (0.001 sec)

```
mysql> SELECT *
-> FROM class
-> RIGHT OUTER JOIN class_info
[ -> ON class.id = class_info.id;
```

| id   | name | id | address |
|------|------|----|---------|
| 1    | abhi | 1  | DELHI   |
| 2    | adam | 2  | MUMBAI  |
| 3    | alex | 3  | CHENNAI |
| NULL | NULL | 7  | NOIDA   |
| NULL | NULL | 8  | PANIPAT |

5 rows in set (0.001 sec)

```
mysql> SELECT class_info.id, class_info.address
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id
[ -> WHERE class.id IS NULL;
```

| id | address |
|----|---------|
| 7  | NOIDA   |
| 8  | PANIPAT |

2 rows in set (0.001 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id;
```

| id   | name   | id   | address |
|------|--------|------|---------|
| 1    | abhi   | 1    | DELHI   |
| 2    | adam   | 2    | MUMBAI  |
| 3    | alex   | 3    | CHENNAI |
| 4    | anu    | NULL | NULL    |
| 5    | ashish | NULL | NULL    |
| NULL | NULL   | 7    | NOIDA   |
| NULL | NULL   | 8    | PANIPAT |

7 rows in set (0.003 sec)

```
mysql> SELECT class.id, class.name, class_info.id, class_info.address
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class_info.id IS NULL
->
-> UNION
->
-> SELECT class.id, class.name, class_info.id, class_info.address
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class.id IS NULL;
```

| id   | name   | id   | address |
|------|--------|------|---------|
| 4    | anu    | NULL | NULL    |
| 5    | ashish | NULL | NULL    |
| NULL | NULL   | 7    | NOIDA   |
| NULL | NULL   | 8    | PANIPAT |

4 rows in set (0.001 sec)

```
mysql> CREATE TABLE first_table (
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.009 sec)
```

```
mysql> CREATE TABLE second_table (
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.014 sec)
```

```
mysql> INSERT INTO first_table VALUES
    -> (1,'abhi'),
[    -> (2,'adam');
Query OK, 2 rows affected (0.003 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO second_table VALUES
    -> (2,'adam'),
[    -> (3,'chester');
Query OK, 2 rows affected (0.003 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
[mysql> SELECT * FROM first_table;
+-----+-----+
| id  | name |
+-----+-----+
|  1  | abhi |
|  2  | adam |
+-----+-----+
2 rows in set (0.000 sec)
```

```
[mysql> SELECT * FROM second_table;
+-----+-----+
| id  | name  |
+-----+-----+
|  2  | adam  |
|  3  | chester |
+-----+-----+
2 rows in set (0.000 sec)
```

```
mysql> SELECT * FROM first_table
    -> UNION
[    -> SELECT * FROM second_table;
+-----+-----+
| id  | name  |
+-----+-----+
|  1  | abhi  |
|  2  | adam  |
|  3  | chester |
+-----+-----+
3 rows in set (0.001 sec)
```

```
mysql> SELECT * FROM first_table
    -> UNION ALL
[    -> SELECT * FROM second_table;
+-----+-----+
| id  | name  |
+-----+-----+
|  1  | abhi  |
|  2  | adam  |
|  2  | adam  |
|  3  | chester |
+-----+-----+
4 rows in set (0.000 sec)
```

```
mysql> SELECT name FROM first_table
-> UNION
[ -> SELECT name FROM second_table;
```

| name    |
|---------|
| abhi    |
| adam    |
| chester |

3 rows in set (0.001 sec)

```
mysql> SELECT name FROM first_table
-> UNION ALL
[ -> SELECT name FROM second_table;
```

| name    |
|---------|
| abhi    |
| adam    |
| adam    |
| chester |

4 rows in set (0.001 sec)

```
mysql> SELECT * FROM first_table
-> INTERSECT
[ -> SELECT * FROM second_table;
```

| id | name |
|----|------|
| 2  | adam |

1 row in set (0.001 sec)

```
[mysql> SELECT name FROM first_table;
```

| name |
|------|
| abhi |
| adam |

2 rows in set (0.000 sec)

```
[mysql> SELECT name FROM second_table;
```

| name    |
|---------|
| adam    |
| chester |

2 rows in set (0.000 sec)



```
[mysql> SELECT name FROM second_table;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| adam |
```

```
| chester |
```

```
+-----+
```

```
2 rows in set (0.000 sec)
```

```
mysql> SELECT name FROM first_table
```

```
-> INTERSECT
```

```
[ -> SELECT name FROM second_table;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| adam |
```

```
+-----+
```

```
1 row in set (0.001 sec)
```

```
mysql> SELECT f.*
```

```
-> FROM first_table f
```

```
-> LEFT JOIN second_table s
```

```
-> ON f.id = s.id
```

```
[ -> WHERE s.id IS NULL;
```

```
+-----+-----+
```

```
| id | name |
```

```
+-----+-----+
```

```
| 1 | abhi |
```

```
+-----+-----+
```

```
1 row in set (0.001 sec)
```

```
mysql> SELECT f.name
```

```
-> FROM first_table f
```

```
-> LEFT JOIN second_table s
```

```
-> ON f.name = s.name
```

```
[ -> WHERE s.name IS NULL;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| abhi |
```

```
+-----+
```

```
1 row in set (0.001 sec)
```

```
[mysql> exit
```

```
Bye
```